Smart IoT-Based Intrusion Detection & Alert System

Project Report

Undertaken at:



Software Technology Park, Guwahati, Assam

by

Shristi Gupta (DC2022BTE0075)

Samriddhi Kumari (DC2022BTE0100)

from

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSAM DON BOSCO UNIVERSITY

AZARA, GUWAHATI 781017,
ASSAM, INDIA.

Acknowledgement

We would like to extend our heartfelt gratitude to our esteemed project guides, **Mr. Lakhajit Hazarika , Mr. Samannay Borah** and **Mr. Gitartha Dev Sarma**, for their unwavering support and encouragement throughout our project journey. Their continuous guidance and expert insights were vital in shaping our understanding and execution of this system. Their mentorship not only enhanced our technical approach but also helped us remain focused and confident during every phase of development.

Under their guidance, we successfully designed and implemented the **Smart IoT-Based Intrusion Detection & Alert System**. This system integrates hardware components such as NodeMCU, Arduino UNO, RFID Module, LM393 Motion Sensor, Servo Motor, and RGB LED to detect unauthorized access and send real-time alerts via the Telegram Bot API. Our mentors provided valuable inputs on optimizing the communication between Arduino and NodeMCU, which was crucial for seamless operation and secure message delivery.

We sincerely thank them for inspiring us to push our boundaries and stay committed to innovation. Their contributions were instrumental in making our project not only functional but also aligned with real-world IoT applications.

Project Members :

- Shristi Gupta (DC2022BTE0075)
- Samriddhi Kumari (DC2022BTE0100)

Declaration

We hereby declare that the project titled **"Smart IoT-Based Intrusion Detection & Alert System"** was successfully submitted to the **Software Technology Park, Guwahati**, under the guidance of **Mr. Lakhajit Hazarika , Mr. Samannay Borah**, and **Mr. Gitartha Dev Sarma**. This project integrates both IoT and microcontroller-based technologies to create a smart, real-time security system capable of detecting intrusions and sending alerts instantly.

The system uses two Arduino UNO boards to handle core tasks such as motion detection, RFID authentication, and door lock control through a servo motor. The NodeMCU (ESP8266) handles wireless communication and sends secure Telegram alerts using the Telegram Bot API. Sensors like the LM393 motion detector and RFID reader (MFRC522) ensure accurate detection and identity verification. Visual feedback is provided via an RGB LED that indicates system states: red for intrusions and green for authorized access.

Communication between the Arduino and NodeMCU is achieved using the SoftwareSerial library, enabling smooth data exchange. The system is modular, fully automated, and designed to provide real-time intrusion alerts and access logs, making it highly effective for secured environments.

Shristi Gupta (DC2022BTE0075)

Department. of Computer Science & Engineering

Samriddhi Kumari (DC2022BTE0100)

Department. of Computer Science & Engineering

Abstract

The **Smart IoT-Based Intrusion Detection & Alert System** is an intelligent and affordable security solution designed to monitor, detect, and respond to unauthorized access in real time. Central to the system is the integration of Arduino Uno and NodeMCU (ESP8266), which collectively manage sensor data and cloud communication. The system utilizes an LM393 motion sensor to detect movement and an SG90 servo motor that simulates a smart locking mechanism. In case of suspicious activity, the system sends real-time notifications via Telegram using the WiFi-enabled NodeMCU, ensuring immediate user awareness.

The project is built with a modular architecture that separates hardware responsibilities: Arduino handles sensor inputs (motion), RFID, RGB LED indication, and servo control, while NodeMCU manages wireless communication and remote alerting through the Telegram Bot API using secure protocols. RFID authentication is implemented using the MFRC522 module, allowing only authorized users to unlock the door. Unauthorized attempts trigger red light alerts and are logged immediately. Communication between Arduino and NodeMCU is enabled through the SoftwareSerial library, allowing seamless data exchange and real-time responsiveness.

This project highlights the potential of IoT in enhancing small-scale surveillance systems using accessible components like jumper wires, breadboards, and open-source software. Visual cues from an RGB LED provide immediate system status: red for alerts, green for authorized access. The fully automated system ensures that every intrusion attempt is immediately detected, responded to, and communicated remotely, making it suitable for home, lab, or office environments.

CONTENT

Sl no.	CONTENT	Page No.
1.	Introduction	1
2.	Problem Definition	1
3.	Objectives	2
4.	Components Used	2
5.	Working Principle	3
6.	Flowchart	4
7.	Features	5
8.	Circuit Diagram	6-8
9.	Code	9-11
10.	Objectives Achieved	12
11.	Future Scope	13
12.	Conclusion	13

1. Introduction

In today's world, securing homes and personal spaces is increasingly important due to rising cases of theft and unauthorized access. Traditional systems often lack real-time responses and smart automation. Our project, **Smart IoT-Based Intrusion Detection & Alert System**, enhances security by integrating IoT components to provide instant and intelligent intrusion detection. It ensures faster communication and automatic responses to potential threats, improving both safety and user awareness.

The system employs two **Arduino UNO boards** and a **NodeMCU (ESP8266)** to manage all hardware and connectivity tasks. The Arduino handles the **RFID module** for authenticating authorized users, the **LM393 motion sensor** for detecting movement, the **SG90 servo motor** for locking/unlocking mechanisms, and an **RGB LED** for visual status indication. NodeMCU connects to Wi-Fi and sends real-time intrusion alerts or access logs through the Telegram Bot API using secure communication.

Data transfer between Arduino and NodeMCU is handled via the SoftwareSerial library. If motion is detected, the RGB turns red, the servo locks, and a Telegram alert is sent. If a valid RFID tag is scanned, access is granted, the servo unlocks, and the RGB turns green. This low-cost, compact system offers reliable, real-time protection for homes, hostel rooms, and offices.

2. Problem Definition

Conventional security systems often rely on manual activation, lack remote alert features, or respond too slowly to potential threats. They usually depend on expensive hardware, CCTV storage, or human monitoring, which may delay reaction time during critical situations. Moreover, these systems may not detect physical tampering or forced access effectively.

This project aims to address these issues by developing an affordable, real-time intrusion detection and response system using IoT technology. By automating detection, alert, and locking processes, the system provides immediate local and remote responses to potential intrusions, enhancing safety and minimizing the chances of unauthorized access.

3. Objectives

- **Detects Intrusion and Lockdown:** An LM393 motion sensor detect unauthorized movement or tampering. Upon detection, the system immediately switches the RGB LED to red for visual alert and locks the servo-controlled door.

- **Authenticate with RFID & Respond:** Arduino verifies user identity using the MFRC522 RFID module. If authorized, the servo unlocks and the RGB LED turns green. Unauthorized access triggers alert mode with red LED and logs the event.

- **Send Real-Time Alerts via Telegram:** NodeMCU uses secure Wi-Fi (ESP-12E) to send instant intrusion or access alerts to the user through Telegram, ensuring real-time monitoring and response from any location.

4. Components Used

Component	Specifications	Purpose
NodeMCU (ESP8266)	Wi-Fi enabled microcontroller (ESP-12E)	Sends real-time alerts via Telegram bot over Wi-Fi
Arduino UNO (2x)	ATmega328P-based microcontroller	Core control unit managing operations
RFID Module (MFRC522)	13.56 MHz RFID reader module	Authenticates authorized personnel via RFID tags
Servo Motor	SG90, 90° rotation	Simulates door lock mechanism (lock/unlock)
LM393 Motion Sensor	Infrared motion detector	Detects unauthorized physical movement
RGB LED	Common cathode, 3 PWM inputs	Indicates system state (Red: Alert, Green: Access Granted)
Jumper Wires	Standard prototyping components	Electrical interconnections
Telegram Bot API	Online messaging service	Sends intrusion alerts and RFID logs to the user

Table 4.1: Components Used

5. Working Principle

- **Motion Detection**: The **LM393 motion sensor** continuously monitors the area. When unauthorized motion is detected, it triggers the **Arduino UNO** to initiate a security check.
- **RFID Authentication**: The **MFRC522 RFID module** scans for valid RFID tags. If an authorized UID is detected, access is granted; otherwise, the system remains in alert mode.
- **Auto-Lock Response**: Upon motion detection or failed RFID authentication, the **SG90 servo motor** rotates to lock the door, simulating a physical security barrier.
- **Visual Alerts**: The **RGB LED** provides instant status feedback — **Red** for intrusion alert, **Green** for access granted.
- **Real-Time Notification**: The **NodeMCU (ESP8266)** sends instant alerts via **Telegram Bot API** over secure communication, keeping the user informed remotely.
- **System Coordination**:Communication between **Arduino UNO** and **NodeMCU** is handled via SoftwareSerial, ensuring seamless data flow.
- **Reliable & Automated**: The system minimizes false alarms by requiring both motion and tamper detection or failed authentication to trigger alerts.

6. Flowchart

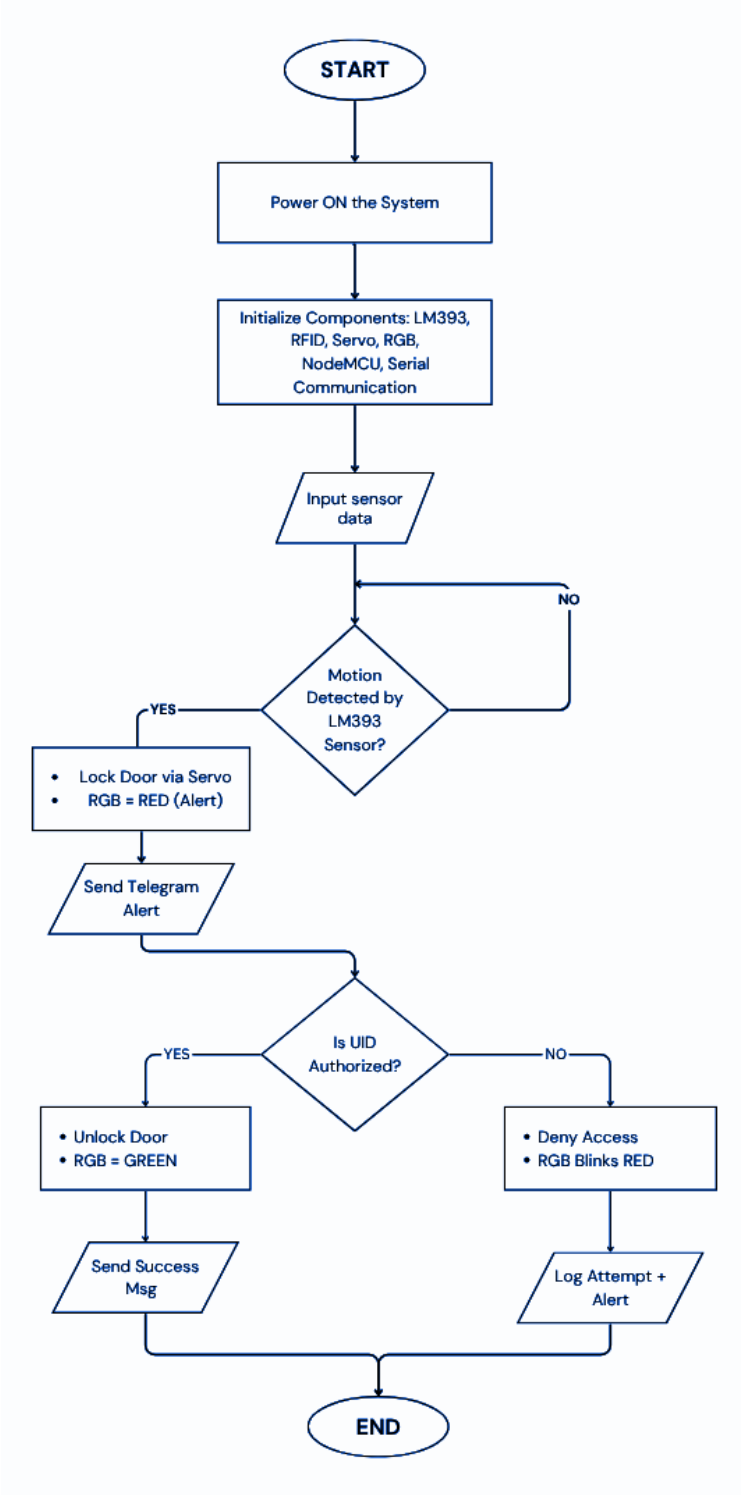


Fig 6.1: Flowchart of the system

7. Features

- Dual-Controller Design: Arduino UNO handles sensors & RFID; NodeMCU manages Wi-Fi and Telegram alerts.
- Real-Time Intrusion Alerts: NodeMCU sends immediate notifications to the user via Telegram using secure HTTPS.
- Dual-Sensor Validation: Motion sensor (LM393) detects movement; RFID authenticates entry.
- Automatic Smart Locking: Servo motor locks door upon motion detection; unlocks only after valid RFID authentication.
- Visual Feedback via RGB LED:
    - Red → Intruder detected or access denied
    - Green → Access granted
- Fully Automated Access Control: Access is granted or denied based on stored UID; no manual intervention needed.
- Modular and Scalable: Components work independently; easy to modify or upgrade.
- Compact & Cost-Effective: Uses off-the-shelf components, suitable for homes, hostels, and small offices.
- Secure Communication: WiFiClientSecure ensures safe Telegram alert delivery.
- Clean Wiring & Prototyping: Uses jumper wires, breadboard, and SoftwareSerial for inter-device communication.

8. Circuit Diagram

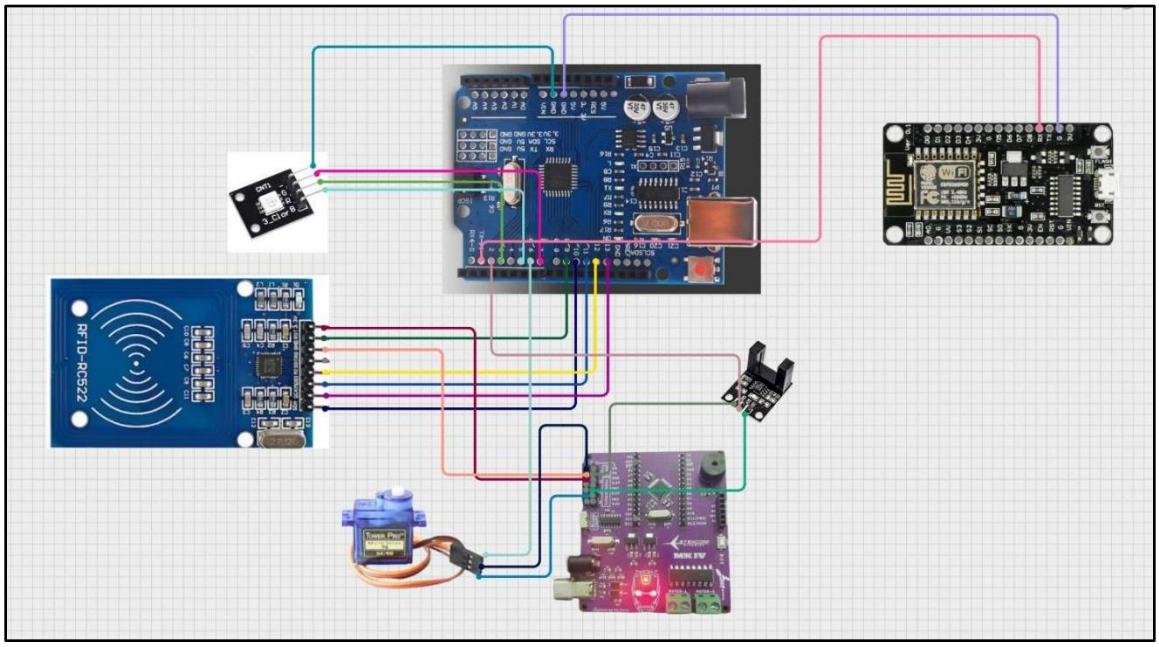
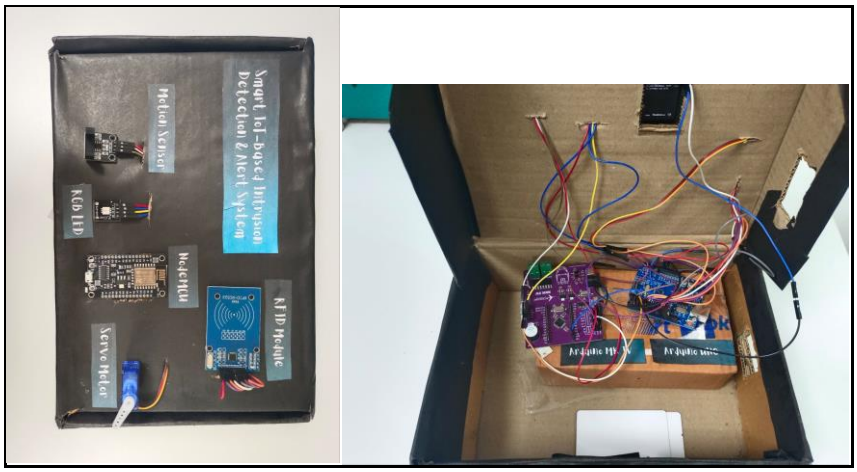


Fig 8.1: Circuit Diagram

Circuit Photographs



(a) (b)

Fig 8.2: (a) Circuit set-up ; (b) Circuit wiring

Arduino UNO:



The Arduino Uno R3 Pro is a microcontroller board based on the ATmega328P. It features 14 digital input/output pins (6 of which can be used as PWM outputs), 6 analog inputs, a USB connection for programming, and a power jack for external power supply. This board is widely used for building interactive projects, prototypes, and educational purposes due to its simplicity, versatility, and extensive community support.

MK IV



The MK IV is an Arduino-based development board created by Stemcorp Dynamics. It's designed for makers, students, and innovators, offering a platform for projects in STEM, robotics, and IoT. The MK4 is Arduino-compatible and features a flexible, modular design. It's also known for being affordable and easy to use.

Esp-8266



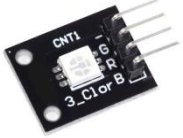
The **ESP8266 12-E** is a low-cost Wi-Fi microchip developed by ESP8266. It integrates a full TCP/IP stack and microcontroller functionality, making it a versatile solution for Internet of Things (IoT) applications. This module is widely used for projects requiring wireless connectivity, such as smart home devices, remote sensors, and automation systems. Its compact size, low power consumption, and robust feature set make it a popular choice among hobbyists and professionals alike.

Motion sensor



This is an **IR slot sensor (LM393),** commonly used for object or motion detection. It works by detecting interruptions in the infrared light between its emitter and receiver. When an object passes through the slot, the sensor outputs a signal, making it useful in applications like motor speed detection and automation systems.

RGB



The KY-009 is a simple RGB LED module manufactured by Arduino (Part ID: 2). This module features a Red-Green-Blue (RGB) LED that allows users to control the color and brightness of the light using Pulse Width Modulation (PWM). It is a versatile component commonly used in projects requiring visual indicators, decorative lighting, or color-based feedback systems.

RFID



RFID uses electromagnetic fields to identify and track tagged objects. It includes a reader and tags, which can be passive or active. Common applications include inventory management, access control, and contactless payments. This guide covers the RC522 RFID module and its integration with microcontrollers like the Arduino UNO.

Servo motor



The SG90 Servo Motor is a compact, lightweight, and affordable servo ideal for robotics and hobby projects. It offers precise angular control using PWM signals and works well with microcontrollers like Arduino. Common uses include robotic arms, camera mounts, and model planes, making it great for beginners and experts alike.

9. Codes

NodeMCU (ESP8266) Code

sketch_aug5b.ino

```ino
#include <ESP8266WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>

// WiFi credentials
const char* ssid = "Shristi";
const char* password = "30112003";

// Telegram Bot details
#define BOT_TOKEN "8394764114:AAGZBfxlJUhdJYmpOTPDPvvnT9JgTYv83nU"
#define CHAT_ID "1722103247"

WiFiClientSecure secured_client;
UniversalTelegramBot bot(BOT_TOKEN, secured_client);

String incomingMessage = "";
unsigned long lastTime = 0;

void setup() {
  Serial.begin(9600);

  WiFi.mode(WIFI_STA);
  WiFi.begin(ssid, password);
  secured_client.setInsecure(); // Disable certificate verification

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\n✅ WiFi Connected");
  bot.sendMessage(CHAT_ID, "🤖 NodeMCU is online and ready!", "");
}

void loop() {
  while (Serial.available()) {
    char c = Serial.read();
    if (c == '\n') {
      bot.sendMessage(CHAT_ID, incomingMessage, "");
      incomingMessage = "";
    } else {
      incomingMessage += c;
    }
  }
}
```

Arduino UNO Code

sketch_aug5a.ino

```
#include <SPI.h>
#include <MFRC522.h>
#include <Servo.h>
#include <SoftwareSerial.h>

// === RFID Pins ===
#define RFID_SDA  10
#define RFID_RST  9

MFRC522 rfid(RFID_SDA, RFID_RST);
Servo myServo;

// === Other Pins ===
#define SENSOR_PIN 2
#define RED_PIN 3
#define GREEN_PIN 7
#define BLUE_PIN 5
#define SERVO_PIN 6

bool systemLocked = false;
String authorizedUID = "E7 10 6E 05";

SoftwareSerial nodeSerial(7, 8); // RX, TX (To NodeMCU)

void setup() {
  Serial.begin(9600);
  nodeSerial.begin(9600);
  SPI.begin();
  rfid.PCD_Init();

  pinMode(SENSOR_PIN, INPUT);
  pinMode(RED_PIN, OUTPUT);
  pinMode(GREEN_PIN, OUTPUT);
  pinMode(BLUE_PIN, OUTPUT);

  myServo.attach(SERVO_PIN);
  myServo.write(0); // Unlock

  Serial.println("System Ready");
  nodeSerial.println("🔧 System Ready. Waiting for motion...");
}
```

sketch_aug5a.ino

```
43  void loop() {
44    if (!systemLocked && digitalRead(SENSOR_PIN) == HIGH) {
45      Serial.println("🚨 Motion Detected");
46      nodeSerial.println("🚨 Motion Detected");
47      setRGBColor(255, 0, 0);
48      myServo.write(90);
49      Serial.println("🔒 Locked");
50      nodeSerial.println("🔒 Servo moved to 90° (Locked)");
51      systemLocked = true;
52    }
53
54    if (systemLocked && rfid.PICC_IsNewCardPresent() && rfid.PICC_ReadCardSerial()) {
55      String uid = getUID();
56      Serial.print("💳 RFID UID: ");
57      Serial.println(uid);
58      nodeSerial.print("💳 RFID UID: ");
59      nodeSerial.println(uid);
60
61      if (uid == authorizedUID) {
62        Serial.println("✅ Access Granted");
63        nodeSerial.println("✅ Access Granted");
64        myServo.write(0);
65        setRGBColor(0, 255, 0);
66        delay(3000);
67        setRGBColor(0, 0, 0);
68        systemLocked = false;
69        Serial.println("🔄 System reset");
70        nodeSerial.println("🔄 System reset. Waiting for next motion...");
71      } else {
72        Serial.println("❌ Access Denied");
73        nodeSerial.println("❌ Access Denied");
74        setRGBColor(255, 0, 0);
75        delay(500);
76        setRGBColor(0, 0, 0);
77      }
78
79      rfid.PICC_HaltA();
80      rfid.PCD_StopCrypto1();
81    }
82  }
```

```
83
84  String getUID() {
85    String content = "";
86    for (byte i = 0; i < rfid.uid.size; i++) {
87      content += (rfid.uid.uidByte[i] < 0x10 ? "0" : "");
88      content += String(rfid.uid.uidByte[i], HEX);
89      if (i < rfid.uid.size - 1) content += " ";
90    }
91    content.toUpperCase();
92    return content;
93  }
94
95  void setRGBColor(int r, int g, int b) {
96    analogWrite(RED_PIN, r);
97    analogWrite(GREEN_PIN, g);
98    analogWrite(BLUE_PIN, b);
99  }
100
```

10. Objectives Achieved

Objective	Description
Detects Intrusion and Lockdown	Uses LM393 motion sensor to detect unauthorized movement. Upon detection, RGB LED turns red and servo-controlled door locks automatically.
Authenticate with RFID & Respond	Arduino reads RFID UID via MFRC522. If UID is authorized, servo unlocks and LED turns green. Unauthorized UIDs trigger red LED and system logs the access.
Send Real-Time Alerts via Telegram	NodeMCU (ESP-12E) securely connects to Wi-Fi and sends real-time intrusion and access alerts via Telegram bot for remote monitoring and response.

Table 10.1 : Objectives Achieved

Telegram Alerts

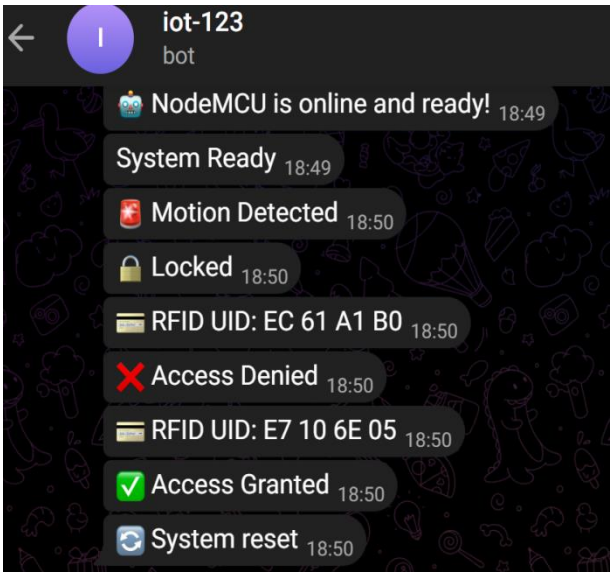


Fig 10.1: Telegram Alerts in Real-time .

11. Future Scope

In future iterations, this system can be enhanced with camera modules capable of capturing images or videos during intrusion events. This addition will provide visual evidence, enabling more accurate verification and helping in investigations. Advanced AI models can also be integrated to differentiate between humans and pets, minimizing false alarms and improving overall system intelligence. The system could further evolve to recognize authorized faces or gestures using computer vision, enhancing automation and security.

Moreover, cloud-based storage can be implemented to maintain secure logs of all intrusion attempts and RFID authentications, accessible from anywhere. Mobile apps may be developed for remote control and monitoring of the system, giving users full access to real-time updates and settings adjustments. Integration with smart voice assistants like Alexa or Google Assistant will make the system more interactive and user-friendly. In addition, solar panel support and a battery backup feature will ensure uninterrupted operation in rural or remote areas where electricity is unreliable. These improvements will make the system not only smarter but also more robust, sustainable, and suitable for a wider range of deployment scenarios.

12. Conclusion

The **Smart IoT-Based Intrusion Detection & Alert System** is a practical blend of IoT innovation and physical security tailored for modern needs. It intelligently combines components such as the Arduino UNO and NodeMCU to manage sensor input, communication, and real-time alert delivery through Telegram. RFID authentication adds an extra layer of access control, while servo motors simulate physical locking mechanisms. The RGB LED provides immediate visual feedback on system status, making it user-friendly and intuitive.

This system proves to be affordable, scalable, and easily deployable in homes, schools, or small offices. With modular hardware and straightforward software logic, it bridges the gap between conventional security systems and modern IoT-based solutions. Its real-time response to motion and unauthorized access ensures proactive security. By enabling automation, remote monitoring, and immediate alerts, this project stands as a significant step toward accessible and efficient personal security for everyone. It successfully delivers on its goal of blending cost-effectiveness with smart functionality.